

TE Computer (V)

Sub: Theory of Computation

Date: 22th Sept 2025

CO4	Design Context free grammar, pushdown automata as acceptor, recognizer of the language.
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Assignment No 03

Topic: Pushdown Automata (PDA)

1. Define PDA and differentiate between NPDA and PDA. Explain NPDA with example.
2. Write a short note on Applications of PDA.
3. Design PDA for the following

$$L = \{a^m b^n \mid n < m\}$$

$$L = \{a^n b^n c^m d^m \mid n, m \geq 1\}$$

$$L = \{a^n b^m a^n \mid n, m \geq 1\}$$
4. Design PDA for even palindrome over $\{a, b\}$
5. Write short note on acceptance by PDA
6. Construct PDA accepting the language, $L = \{a^{2n} b^n \mid n \geq 0\}$